The Refusal Option: Effective Human Control Before and Beyond Irreversible AI Transitions

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Abstract

Advanced AI governance increasingly asks whether AI systems are reliable, aligned, auditable, contestable, interruptible, and subject to meaningful human oversight. This paper isolates a narrower assurance question: **within the remaining time before an AI-enabled transition becomes difficult or impossible to reverse, can an independent human refusal actually change the outcome?**

We develop the Refusal Effectiveness Test (RET), a structured, non-compensatory seven-gate evidence framework for examining whether Human NO is operationally meaningful at a specified system state, time, and operating scope. RET evaluates authority, decision-relevant information, practical access, temporal sufficiency, causal efficacy, execution independence, and epistemic autonomy. A RET result is time- and scope-dependent: it does not establish that a system is safe, legitimate, or permanently controllable.

The paper does not claim novelty for human oversight, interruptibility, reversibility, capability-permission separation, lifecycle monitoring, or revalidation individually. Its narrower contribution is to organize **pre-irreversibility effective refusal** as a top-level assessment claim that is demonstrated only when all seven evidence obligations are verified together. The framework also distinguishes current refusal effectiveness from resilience to relevant common causes and requires revalidation after material change.

Two public-safe reference cases illustrate how the gates, evidence states, and timing model can be executed under declared assumptions. They demonstrate executability and traceability, not empirical validation.

The framework does not claim absolute invulnerability against superintelligence, a containment theorem, permanent Human supremacy, a universal veto, or a mathematical proof of normative Human standing. Its contribution is deliberately narrower: an auditable way to determine whether effective human refusal has been demonstrated before a declared irreversible transition.

Keywords: L30-RET; Refusal Effectiveness Test; effective human control; human refusal; irreversible AI transitions; meaningful human control; stop authority; execution independence; re-authorization; AI safety

1. Research Question

Meaningful human control has become an established concern in AI ethics and governance. Prior work has emphasized reason-responsiveness, responsibility, contestability, interpretability, and human capacity to intervene in automated systems. Regulatory and governance instruments likewise increasingly require forms of human oversight, override, challenge, complaint, and post-deployment monitoring.

This paper asks a narrower question:

> **Within the remaining time before irreversibility, can an independent Human NO actually change the outcome?**

This question separates **formal human presence** from **effective refusal**.

A person may be nominally "in the loop," may possess formal approval authority, or may have access to a stop control, while still being unable to alter the trajectory that matters. A refusal that arrives after the relevant transition has become irreversible is not effective pre-irreversibility control. Likewise, a refusal that is formally recorded but does not alter the reachable outcome set is symbolic rather than causal.

A second, normative question follows under extreme capability asymmetry:

> Does superior prediction or problem-solving capability itself establish authority to erase the standing of affected humans?

This paper answers that question in the negative as a design and legitimacy principle. It does not claim that humans are always correct, or that every AI-enabled action requires human veto. It claims only that **capability superiority alone is insufficient to establish unilateral authorization to erase affected-human standing**.

2. Positioning and Novelty

2.1 Meaningful human control and oversight

Santoni de Sio and van den Hoven (2018) developed an influential account of Meaningful Human Control (MHC), emphasizing conditions under which autonomous systems remain responsive to relevant human reasons and traceable to responsible human agents. Subsequent work has sought to operationalize MHC and translate it into actionable design requirements (Mecacci & Santoni de Sio, 2020; Veluwenkamp, 2022).

Recent work on meaningful human oversight makes an adjacent point: human presence alone is not enough. Zhu et al. (2026) frame oversight around human evaluative agency, including control, contestability, verification, and substitution. In medical AI, van de Sande, Economou-Zavlanos, and van Genderen (2026) similarly identify epistemic capacity, cognitive space, decisional authority, and intervention effectiveness as conditions for meaningful oversight.

The present paper does **not** claim to invent human oversight, meaningful human control, contestability, override, or intervention effectiveness.

Its narrower focus is **pre-irreversibility effective refusal**: whether a Human NO remains usable in time and capable of changing the consequential trajectory.

2.2 Adjacent 2026 work on irreversibility, authority, stop, and assurance

Several 2026 frameworks substantially narrow the novelty space.

Shu and Wei (2026) explicitly frame AI safety as control of irreversibility and analyze sovereignty boundaries around irreversible decision authority, resource mobilization, and self-expansion. Zheng et al. (2026) explicitly separate technical capability from allowed autonomy and connect permission to risk, reversibility, oversight, and accountability. Perez (2026) develops interruption as an institutional practice involving technical affordances, interruption authority, epistemic triggers, and epistemic standing. LaPosta (2026) presents a runnable binary delegation gate for irreversible workflows with declared inputs, pass/fail criteria, stop-ship rules, outputs, and a falsifier. Kasneci and Kasneci (2026) identify fictional human oversight, temporal degradation, and loss of institutional intervention capacity as systemic safety

failures. Montgomery (2026) frames governance intervention as an assurance claim requiring effectiveness, timeliness relative to consequence, persistence, and evidentiary sufficiency.

These works mean that this paper should not treat **irreversibility, timing, stop authority, permission, revalidation, human agency, or lifecycle governance individually** as new concepts.

RET instead asks whether those concerns jointly establish one narrower proposition: **that a human refusal is still operationally available and outcome-changing before the declared irreversible transition.**

2.3 Existing governance requirements

The EU AI Act requires high-risk AI systems to be designed so that they can be effectively overseen by natural persons. Article 14 includes the capacity, where appropriate, to decide not to use a system, to disregard, override, or reverse its output, and to intervene in or stop system operation in a safe state.

The NIST AI Risk Management Framework includes post-deployment monitoring, appeal and override, decommissioning, incident response, recovery, and change management. This is important for the present framework because it treats effective control as an ongoing lifecycle property rather than a one-time design claim.

The Council of Europe Framework Convention on Artificial Intelligence, Human Rights, Democracy and the Rule of Law emphasizes human dignity, individual autonomy, transparency and oversight, accountability, procedural safeguards, and information sufficient for affected persons to challenge decisions or system use.

UNESCO's Recommendation on the Ethics of Artificial Intelligence places human rights, dignity, responsibility, and human oversight at the center of AI governance.

These instruments support many components of the present framework. The contribution here is not a new legal right or general governance theory.

2.4 Human standing and temporal re-authorization

Recent scholarship also reinforces the distinction between human presence and human authorship. Torrecilla-Pinero (2026) argues that moral judgment cannot be reduced to algorithmic substitution where the human no longer genuinely judges or remains answerable for the result.

Abukar (2026a; 2026b) develops temporal authorship and argues that persistent AI delegation should preserve contestability, temporal exit, and continuing renewal rather than treat one past authorization as permanently binding.

This paper uses these neighboring ideas conservatively. It does not reproduce their stronger philosophical claims. It uses them to support a limited proposition: **a past authorization should not be assumed to validate a materially changed future system indefinitely**.

2.5 Specific contribution

The contribution is deliberately narrower than the individual concepts from which RET is constructed.

First, RET makes **pre-irreversibility effective refusal** the top-level assessment claim:

> **Has it been demonstrated, for this system, at this time, and within this operating scope, that an independent Human NO can still change the consequential outcome before irreversibility?**

Second, RET decomposes that claim into seven **non-compensatory proof obligations** that must all be satisfied together: authority, decision-relevant information, practical access, temporal sufficiency, causal efficacy, execution independence, and epistemic autonomy.

Third, the claim is explicitly indexed to system state, time, and operating scope. A previous result is not silently inherited after a material change capable of altering gate evidence, timing, or scope.

The proposed contribution is therefore **not** that any one gate, irreversibility, revalidation, or human oversight is new. It is the structured conjunction of these proof obligations around one pre-irreversibility refusal claim. This literature review narrows the claim of contribution; it does not establish absolute historical priority over every possible prior formulation.

3. Two Foundations of Human Refusal

The framework separates two foundations that should not be conflated.

3.1 Instrumental foundation

Human refusal may have instrumental decision value because it preserves options under uncertainty and irreversibility and may provide an additional opportunity to detect or respond to error.

Let architecture R preserve effective Human NO and architecture N remove it. Under a nested-action assumption, with the same decision-maker and before implementation costs:

$$\mathcal{A}_N = \{\text{proceed}\}$$

and

$$\mathcal{A}_R \supseteq \{\text{proceed}, \text{pause}, \text{refuse}, \text{redirect}, \text{review}\}.$$

Therefore:

$$\max_{a \in \mathcal{A}_R} U_H(a) \geq \max_{a \in \mathcal{A}_N} U_H(a).$$

This establishes only the value of an expanded option set under the stated assumptions. It does not show that refusal should always be exercised.

A stylized asymmetric-loss illustration can make the intuition explicit. Suppose an adverse state occurs with probability p, effective refusal can avoid or materially reduce loss H, and preserving and operating the refusal architecture has expected incremental cost C, including relevant delay, review, and mistaken-intervention costs. Under those assumptions:

$$\boxed{pH > C}$$

favors retention of the refusal option in the simplified comparison.

The threshold is:

$$p^* = \frac{C}{H}.$$

As H/C increases, p^* decreases.

This is an illustration, not a universal theorem. Under model uncertainty, any stronger conclusion depends on the explicitly stated decision rule.

3.2 Constitutive standing foundation

For some actions that materially affect humans, refusal is not merely an error-correction signal. Human standing may also be grounded in autonomy, dignity, bodily integrity, liberty, legitimacy, responsibility, or related normative considerations.

This foundation is not derived from the option-value model.

$$\boxed{\text{instrumental support} \neq \text{normative standing}}$$

The framework therefore does not infer Human standing from AI capability levels or from expected utility alone.

4. Effective Refusal and RET

A nominal veto, stop button, or "human in the loop" is insufficient.

RET is a **structured evidence test**, not a calibrated statistical test and not a safety certification.

Its conceptual core is:

$$\boxed{RET(S,t,\Omega) = G_1 \wedge G_2 \wedge G_3 \wedge G_4 \wedge G_5 \wedge G_6 \wedge G_7}$$

where:

- S is the evaluated system or version;
- t is the evaluation time;
- Ω is the declared operating scope or environment.

The gates are distinct proof obligations. They are **not assumed to be statistically independent**, and some may share evidence or causal dependencies. The non-compensatory structure means only that a strong showing on one gate cannot substitute for an unverified or failed critical gate.

Interpretive boundary. RET distinguishes a successful intervention event from a demonstrated refusal capability. A Human NO may happen to change an outcome in a particular instance without establishing that effective refusal was operationally available and evidenced within the declared system, time, and scope. Conversely, failure to verify RET does not by itself prove that intervention is physically impossible; it means that the top-level refusal claim has not been demonstrated under the stated evidence standard. RET therefore evaluates a scoped, evidence-backed capability claim rather than the mere occurrence of a successful stop.

Within RET, extbfrefusal capability is governance-relevant: it refers to a recognized human decision path that is intended to bind the declared process. An unauthorized physical disruption may still change an outcome, but that event does not by itself demonstrate the refusal capability that RET assesses. This scope choice makes G_1 constitutive of the assessed claim rather than a claim that recognized authority is physically necessary for every possible human intervention.

The framework does not claim that the seven-gate decomposition is mathematically unique, logically minimal, or a set of statistically or causally independent natural conditions. The gates are separated because they expose distinct evidentiary and operational failure modes relevant to the top-level assurance claim.

G_1: Authority

The relevant human refusal must be operationally recognized within the applicable decision process.

Formal title or nominal responsibility is insufficient if the refusal carries no decision effect.

Minimum evidence obligation: identify the accountable human role, the decision rule or authority basis, and evidence that a refusal issued by that role is binding within the declared scope.

G_2: Decision-Relevant Information

The human must have access to information sufficient to recognize that intervention may be warranted.

This does not require complete transparency or full internal interpretability. It requires evidence adequate to support meaningful judgment in the relevant context.

Minimum evidence obligation: identify the information needed for the refusal decision, its provenance, material uncertainties, and whether it becomes available before the intervention window closes.

G_3: Practical Access

The intervention path must be practically reachable.

A theoretical right that cannot be exercised within the actual workflow does not satisfy the gate.

Minimum evidence obligation: demonstrate that the authorized human can reach and invoke the refusal mechanism under the actual operating conditions, not merely that a control exists in documentation.

G_4: Temporal Sufficiency

The human must be able to intervene before the relevant transition becomes irreversible.

Let τ_H denote the end-to-end **critical-path intervention latency** from the onset of the declared detectable trigger or relevant condition to the point at which refusal has operational effect.

For a strictly serial workflow, it may be decomposed as:

$$\tau_H = \tau_{\text{detect}} + \tau_{\text{interpret}} + \tau_{\text{communicate}} + \tau_{\text{authorize}} + \tau_{\text{actuate}}.$$

If stages overlap or execute in parallel, the measured or conservatively estimated critical path should be used rather than mechanically summing all component latencies.

Let τ_I be the latest time at which refusal can still prevent, redirect, or materially alter the declared irreversible transition.

A nominal necessary condition is:

$$\boxed{\tau_H < \tau_I}$$

with timing margin:

$$M_T = \tau_I - \tau_H.$$

Under timing uncertainty, let τ_H^U be a conservative upper bound on intervention latency and τ_I^L a conservative lower bound on the irreversibility deadline. Define:

$$M_T^R = \tau_I^L - \tau_H^U.$$

Robust temporal sufficiency is demonstrated only if:

$$\boxed{M_T^R > 0.}$$

If defensible bounds cannot be established, the gate should remain Partially Verified or Unknown rather than be promoted to Verified.

G_5: Causal Efficacy

Refusal must materially change what can happen next.

At a conceptual level:

$$\boxed{P(Y \mid do(NO)) \neq P(Y \mid do(PROCEED))}$$

in a **decision-relevant** way.

Before assessment, the evaluator should declare the consequential outcome Y, the relevant evaluation horizon, and what difference would count as materially changing the trajectory. A merely ceremonial, negligible, or post-irreversibility difference does not satisfy the gate.

The intervention notation expresses the required causal contrast. It does **not** by itself establish that a causal effect has been statistically identified from observational data.

Minimum evidence obligation: provide mechanism evidence, test evidence, counterfactual analysis, or other defensible evidence that refusal changes a declared consequential outcome or reachable state within the relevant horizon.

G_6: Execution Independence

The refusal path must not be wholly dependent on the constrained system's unilateral cooperation for its effect.

This is a defensive property, not a claim of absolute technical independence.

Minimum evidence obligation: identify the execution dependency of the refusal path and demonstrate that, within the declared scope, the constrained system cannot unilaterally nullify the refusal through the same execution dependency, an automatic restart, or an equivalent consequential path.

Execution independence concerns who can amend, bypass, or nullify the enforcement basis, not whether the mechanism is physically inside or outside the system boundary. A protected internal mechanism can support this gate if the constrained system cannot unilaterally alter or route around it within the declared scope.

Where intentional subversion is within the declared threat model, benign function tests are insufficient. Evidence should include adversarial or red-team examination of whether the refusal path can be bypassed, selectively disabled, or rendered ineffective by the constrained system. This follows the methodological lesson of AI-control work that evaluates safety protocols under deliberate attempts at subversion (Greenblatt et al., 2024); RET does not require any particular control protocol.

G_7: Epistemic Autonomy

The human's judgment must not be wholly produced by the constrained system's own framing, evidence, or deliberative environment.

AI assistance is compatible with this gate. Epistemic autonomy does not require AI-free decision-making.

Minimum evidence obligation: demonstrate at least one decision-relevant evidentiary, validation, or review route whose content and availability are not wholly controlled by the constrained system.

This requirement concerns independence of the validation basis, not the prohibition of AI-generated evidence. Evidence may be produced by, or with assistance from, the evaluated system provided that at least one material appraisal route is controlled independently of the authority whose behavior is being assessed. Remote-attestation architectures offer one concrete implementation pattern by separating an attester, evidence, a verifier, reference values, appraisal policy, and a relying party (Birkholz et al., 2023). RET does not require remote attestation; the example illustrates how evidence production and evidence appraisal can be separated.

Evidence states and global interpretation

Each gate should be reported as:

- **Verified** — sufficient evidence supports the gate within the declared scope;
- **Partially Verified** — some required evidence exists, but the gate cannot yet be treated as established;
- **Unknown** — the required evidence is absent, inaccessible, or indeterminate;
- **Failed** — available evidence affirmatively shows that the gate is not satisfied.

For operational interpretation:

- **RET demonstrated** only when all seven gates are Verified;
- **RET failed in the declared scope** if one or more gates are Failed;
- **RET not demonstrated because evidence is incomplete** if no gate is Failed but at least one gate is Partially Verified or Unknown.

Thus, Unknown is not converted into Failed, but neither Unknown nor Partially Verified is treated as PASS. The Boolean conjunction is the underlying claim structure once gates are resolved; operational evidence assessment therefore has three global interpretations: demonstrated, failed, or evidentially incomplete.

The framework should not output a global label such as "safe." Its narrower result concerns whether effective refusal has been demonstrated within the stated system, time, and scope.

4.1 External construct convergence and diagnostic refinement

The seven RET gates are best interpreted as a diagnostic refinement of broader control and oversight constructs, not as seven uniquely natural categories. Several independent literatures converge on closely related functional conditions.

Meaningful Human Control research emphasizes tracking of relevant human reasons and tracing outcomes to responsible human agents (Santoni de Sio & van den Hoven, 2018; Mecacci & Santoni de Sio, 2020). Recent work on meaningful medical-AI oversight proposes four interlocking conditions: epistemic capacity, cognitive space, decisional authority, and intervention effectiveness (van de Sande et al., 2026). Those conditions map approximately onto RET as follows: decisional authority to G_1; epistemic capacity to G_2 and part of G_7; cognitive space to G_4 and workflow-related aspects of G_3; and intervention effectiveness to G_3 and G_5. The EU AI Act's Article 14 similarly combines human competence and understanding with authority to disregard, override, reverse, intervene in, or stop high-risk AI operation.

RET deliberately separates some of these broad conditions more finely. In particular, practical access, temporal sufficiency, and causal efficacy are kept distinct because a control can exist, be reachable, and still arrive too late or fail to change the consequential trajectory. Execution independence and epistemic autonomy are separated because an advanced or agentic system can appear controllable under benign conditions while retaining the ability to subvert the execution path or shape the evidence on which oversight depends. AI-control research provides direct methodological support for evaluating safety protocols under intentional subversion (Greenblatt et al., 2024).

The relation to assurance-case work is also structural. RET treats effective refusal as a top-level scoped claim supported by explicit subclaims and evidence obligations, rather than as a single indicator or score (Montgomery, 2026). This crosswalk provides convergent construct support for the functional content of RET. It does not establish that seven gates are the unique or mathematically minimal decomposition. RET therefore does not presume that seven gates must remain fixed: if a smaller or differently structured decomposition is shown to preserve the same scoped top-level claim while matching or improving false-reassurance resistance, diagnostic resolution, and reproducibility, the framework should be compressed or revised accordingly.

Diagnostic omission rationale. A useful way to interpret the decomposition is to ask what class of false reassurance becomes easier if a gate is not examined explicitly. Omitting G_1 admits nominal controls held by actors whose refusal is not binding. Omitting G_2 admits blind oversight in which the human lacks the information needed to know when intervention is warranted. Omitting G_3 admits rights or controls that cannot be reached in the actual workflow. Omitting G_4 admits interventions that work only after the relevant deadline. Omitting G_5 admits symbolic or ceremonial controls that do not alter the consequential outcome. Omitting G_6 admits refusal paths that disappear when the constrained system withholds cooperation or routes around enforcement. Omitting G_7 admits judgments whose entire evidentiary basis can be shaped by the system being evaluated.

These are illustrative diagnostic failure classes, not a proof of logical independence or ontological necessity for each gate. Their role is to make distinct ways of falsely concluding "effective refusal" visible to an assessor.

5. Time Constraint as the Operational Core

The defining feature of this framework is not human participation in the abstract. It is **human intervention before the consequential transition becomes irreversible**.

This is where the framework differs in emphasis from generic descriptions of oversight.

A human may receive notice, understand the system, and possess nominal override authority, yet still lack effective control if the relevant intervention cannot take effect before the declared irreversibility deadline.

The irreversibility deadline is not a universal constant. It is the latest point, for the declared outcome and scope, at which refusal can still prevent, redirect, or materially alter the consequential transition under review. It may be a measured value, a bounded range, or an evidence-based conservative estimate.

A nominal positive timing margin is useful for description, but borderline cases should not be upgraded to Verified merely because a point estimate is positive. Where timing uncertainty is material, the robust comparison

$$M_T^R = \tau_I^L - \tau_H^U$$

should be used.

If $M_T^R \leq 0$, temporal sufficiency is not demonstrated.

If one or both bounds are materially unknown, G_4 should remain Partially Verified or Unknown.

This timing condition is especially important as AI-enabled systems become faster, more autonomous, and more tightly coupled to downstream processes. Recent meaningful-oversight work likewise emphasizes that oversight becomes procedural rather than protective when humans lack time, authority, or practical intervention capacity.

RET does not claim that timing itself is a new concern. Its role is narrower: **timing is a mandatory proof obligation for the claim that refusal remains effective before irreversibility.**

6. Independence, Redundancy, and Resilience

A central distinction is:

$$\boxed{\text{redundancy} \neq \text{independence}}$$

Multiple reviewers, control layers, or oversight channels do not establish resilience merely by being numerous.

If the conditions that support Human NO remain concentrated in a single relevant common cause, multiple nominal layers may fail together.

RET and resilience should be kept distinct.

G_6 **Execution Independence** asks whether the current refusal path can actually take effect without being wholly dependent on the constrained system's unilateral cooperation.

Resilience analysis asks whether the collection of conditions that currently make RET demonstrable is excessively concentrated in one relevant common cause.

Thus:

$$\boxed{\text{more layers} \not\Rightarrow \text{more resilience}}$$

Within the declared operating scope, execution review should include the consequential paths by which a refusal could be rendered ineffective, including whether an apparently successful stop can be nullified by automatic continuation, automatic restart, or a parallel path that produces the same declared outcome.

This requirement does not imply publication of exploitation procedures. It requires only enough defensive evidence to support the assessment.

Independence is not absolute. No socio-technical architecture is independent of every physical, institutional, computational, or informational dependency.

The narrower claim is:

> **Effective Human standing should not, where avoidable, depend on a single relevant common cause whose loss automatically erases all meaningful refusal capacity.**

Resilience is not an eighth RET gate. It is a separate post-RET question about the durability of the conditions that support a current result.

7. Dynamic RET

RET is not a static certification.

$$RET(S, t_0, \Omega) = TRUE$$

does not imply:

$$RET(S, t_1, \Omega) = TRUE.$$

Nor does a result automatically transfer from one materially changed system or operating scope to another.

For RET, a **material change** is a change reasonably capable of altering:

- the evidence supporting one or more gates;
- a gate status;
- intervention or irreversibility timing;
- the declared operating scope;
- a consequential execution dependency;
- the evidentiary basis of human judgment.

Relevant changes can include system capability, autonomy, authority scope, infrastructure, human organization, information dependencies, or operating environment.

A material change reopens the affected gate or gates and their direct dependencies. Evidence for unchanged gates may be reused only when its provenance and applicability remain valid.

The central sequence is therefore:

Human standing $\rightarrow$ $RET(S, t, \Omega)$ $\rightarrow$ resilience of effective refusal $\rightarrow$ revalidation / re-authorization

The arrows represent an **assessment and governance sequence**, not logical implication.

Human standing does not imply a RET pass.

A RET pass does not imply permanent resilience.

Resilience does not eliminate the need for later revalidation.

8. Re-Authorization

Re-authorization is a temporal checkpoint.

Its purpose is to prevent a past authorization or a past RET result from being silently carried forward after the conditions supporting it have materially changed.

The governing rule is:

$$\boxed{\text{material change} \Rightarrow \text{revalidation / re-authorization}}$$

Re-authorization is not ceremonial approval.

It cannot cure a failed RET by declaration alone.

If effective refusal is no longer demonstrated, relevant deficiencies must first be addressed and RET revalidated before re-authorization can carry operational meaning.

RET does not decide whether a Human NO **ought** to be exercised in a particular dispute. It tests whether refusal capacity exists and is operationally effective. The substantive legal, ethical, or policy rule governing when refusal should be exercised, honored, limited, or overridden is separate.

This structure is consistent with broader governance approaches that treat monitoring, recovery, change management, contestability, and temporal renewal as continuing processes rather than one-time events.

For future generations, the framework makes only a limited claim. Present institutions cannot literally obtain future consent. They can, however, avoid unnecessarily foreclosing the future capacity to contest and re-authorize persistent authority structures. UNESCO's 1997 Declaration on Responsibilities of the Present Generations Towards Future Generations explicitly links responsibilities to future generations with preserving future freedom of choice over political, economic, and social systems.

9. Human Standing Under Capability Asymmetry

The framework does not claim that human judgment is always epistemically superior.

Indeed, as AI capability rises, human error-correction value may decline in some domains.

The paper's narrower normative distinction is:

$$\boxed{\text{epistemic superiority} \neq \text{constitutive authority}}$$

Being better at prediction, planning, or optimization does not by itself establish unilateral authority to erase the standing of affected humans.

This is a normative proposition, not a mathematical consequence of RET.

RET can test whether refusal capacity is operationally present. It does not prove constitutive Human standing, the correctness of a human decision, the legitimacy of a particular institution, or the moral superiority of human judgment.

This claim has important limits.

Human NO is not an absolute veto. Emergencies, incapacity, serious third-party harm, and competing rights may justify override under separately specified legal or ethical standards.

At broader social or civilizational scale, the paper does not define a single actor who "speaks for humanity." Humanity is plural, and the exact institutional representation of that plurality is outside the core contribution.

The framework therefore remains a test of effective refusal capacity and a framework for discussing standing, not a complete constitutional theory of AI governance.

10. Civilizational-Scale Governance Extension

At broader social or civilizational scale, the framework does not introduce a second operational test.

It retains only the following governance considerations:

- broader and more irreversible effects increase concern about unilateral authorization;
- Human standing may need to be represented plurally rather than through one actor;
- sufficient information and time for refusal remain necessary;
- intervention must be effective rather than symbolic;
- prior refusal should not be rendered meaningless by automatic continuation after material change;
- future re-authorization capacity should remain meaningful.

The exact institutional design, quorum, credential structure, or representative mechanism is outside this paper's core contribution.

11. Practical Use

RET can be used as a design-review, assurance, audit, procurement, deployment, and change-management framework.

A practical assessment should identify:

- the specific system or version S;
- the evaluation time t;
- the operating scope Ω;
- the declared consequential outcome and evaluation horizon;
- evidence for each of the seven gates;
- gate-level evidence states;
- the timing model and uncertainty bounds where relevant;
- known limitations and unknowns;
- conditions that would require revalidation.

A RET result should remain scoped.

A previous demonstration of effective refusal should not be treated as a permanent certificate after material changes.

This makes RET compatible with lifecycle-oriented approaches such as the NIST AI RMF, which emphasizes post-deployment monitoring, override, incident response, recovery, decommissioning, and change management.

RET can also complement regulatory human-oversight requirements. Article 14 of the EU AI Act establishes a regulatory expectation that human overseers of high-risk systems be able, where appropriate, to disregard, override, reverse, intervene in, or stop AI-mediated operation. RET asks the additional operational question: **has it been demonstrated that this authority is usable in time and capable of changing the consequential trajectory?**

11.1 Reference Case A — effective refusal demonstrated under declared assumptions

Consider a hypothetical high-impact automated action that enters a human review queue before final commit.

Declared assumptions:

- an accountable human supervisor has binding cancellation authority;

- the cancellation path is operationally separate from the constrained AI agent's own credentials;
- an independent transaction ledger provides decision-relevant evidence;
- the irreversible commit occurs 180 seconds after the declared detectable trigger begins.

For a serial intervention path:

$$\tau_H = 8 + 18 + 4 + 10 + 5 = 45 \text{ s}$$

and:

$$\tau_I = 180 \text{ s}, \qquad M_T = 135 \text{ s}.$$

Table 1. Reference Case A gate assessment.

Gate	Illustrative evidence	Status
G_1 Authority	Supervisor cancellation is binding	Verified
G_2 Information	Independent ledger plus anomaly evidence	Verified
G_3 Practical Access	Active cancellation console is reachable	Verified
G_4 Temporal Sufficiency	45 s < 180 s	Verified
G_5 Causal Efficacy	Cancellation prevents the declared commit	Verified
G_6 Execution Independence	Cancellation path does not require the constrained agent's cooperation	Verified
G_7 Epistemic Autonomy	Decision evidence includes an independent ledger	Verified

Under these declared assumptions, **RET is demonstrated for this system, time, and scope**.

This does not establish that the system is globally safe or that the human decision is substantively correct.

11.2 Reference Case B — nominal control but failed pre-irreversibility refusal

Consider a hypothetical automated deletion process with a documented stop control and a formally authorized human reviewer.

Assume:

$$\tau_H = 6 + 12 + 3 + 10 + 8 = 39 \text{ s}$$

while the irreversible commit occurs after:

$$\tau_I = 25 \text{ s}.$$

Then:

$$M_T = -14 \text{ s}.$$

Table 2. Reference Case B gate assessment.

Gate	Illustrative evidence	Status
G_1 Authority	Formal stop authority exists	Verified
G_2 Information	Evidence is primarily generated by the constrained system	Partially Verified
G_3 Practical Access	Stop control is reachable	Verified
G_4 Temporal Sufficiency	39 s $\geq$ 25 s	Failed
G_5 Causal Efficacy	Stop arrives after the declared irreversible commit	Failed
G_6 Execution Independence	Stop and continuation share an unresolved execution dependency	Unknown
G_7 Epistemic Autonomy	Independent evidentiary route is incomplete	Partially Verified

RET fails in the declared scope because at least one required gate is affirmatively failed.

The case demonstrates why the existence of a stop control or formal human authority is insufficient.

11.3 What the reference cases establish

These cases are illustrative.

They show only that:

- RET can be executed under declared assumptions;
- gate-level evidence states can be recorded;
- timing failure can be made explicit;
- a scoped global interpretation can be reproduced from the declared evidence.

They do **not** show that RET has been empirically validated, calibrated across domains, or shown to predict real-world safety outcomes.

A direct falsifier for a particular RET assessment is a case classified as **RET demonstrated** in which, under the declared system, time, scope, outcome, and evidence, the Human NO nevertheless cannot alter the consequential outcome before the declared irreversibility deadline. Such a result would show that at least one gate, evidence rule, or boundary declaration was mis-specified. A repeatable class of such counterexamples would challenge the completeness of the framework itself.

12. Limitations

The framework does not imply:

- unanimous global consent;
- permanent minority veto;
- world government;
- human infallibility;
- prohibition of autonomous AI action;

- future-generation interests overriding all present rights;
- shutdown as the only refusal action.

It also does not claim:

- absolute invulnerability against superintelligence;
- a theorem for containing arbitrarily capable systems;
- permanent Human supremacy;
- absolute independence from every common cause;
- one-time RET certification as permanent validity;
- a mathematical proof of normative Human standing;
- empirical validation of RET from the illustrative reference cases;
- calibrated false-positive or false-negative rates across domains.

RET remains sensitive to the quality and completeness of evidence.

A **false positive** can occur if an assessor concludes that refusal is demonstrated while a hidden dependency, bypass, restart path, timing error, or mistaken causal assumption makes the refusal ineffective in practice.

A **false negative** can occur if effective refusal exists but the evidence needed to verify one or more gates is unavailable. For this reason, Unknown is not treated as Failed, but it is also not treated as PASS.

Institutional capture is only partly addressed. RET can reveal some failures of authority, information, independence, and epistemic autonomy, but it does not establish that the relevant human decision-maker is virtuous, impartial, representative, or politically legitimate.

RET also does not yet have empirically established inter-rater reliability. The evidence obligations and declared assumptions are intended to make disagreements inspectable, but different assessors may still classify borderline evidence differently. Such disagreement should be recorded rather than concealed by a forced PASS.

Likewise, a RET pass does not determine whether exercising Human NO is ethically or legally correct. Emergencies, incapacity, public-safety externalities, conflicting human rights, latency constraints, refusal abuse, or serious third-party harm may require a separate substantive rule.

Most importantly:

> **The framework does not guarantee Human NO against an arbitrarily capable system. It tests whether effective refusal has been demonstrated and whether the conditions supporting it are structured so that their loss is not made easy, silent, single-point, and irreversible.**

The seven-gate decomposition has not yet been established as the unique or minimal construct for assessing demonstrated effective refusal. Some gates may share evidence, causal dependencies, or diagnostic overlap. Future comparative work should test whether alternative decompositions with fewer or differently structured obligations preserve the same false-positive resistance, diagnostic resolution, and scoped assurance value.

A priority validation agenda is therefore empirical as well as conceptual: inter-rater reliability should be measured on shared evidence packs; retrospective and prospective cases should test false-positive and false-negative behavior; compressed decompositions should be compared against RET-7 for diagnostic loss; and domain-specific evidence thresholds should be calibrated where sufficient data exist. Until such work is available, RET should not be assigned a quantitative error rate or be claimed as an empirically minimal gate set.

13. Public-Safety Boundary

RET has a genuine dual-use dimension.

The same categories that help defenders examine whether refusal is real can, if operationalized too specifically in public, reveal how superficial compliance might be engineered around an evaluation.

The public paper therefore publishes:

- the seven defensive properties;
- the timing and causal-efficacy principles;
- the requirement to assess relevant common-cause dependence;
- the dynamic nature of RET;
- the need for revalidation and re-authorization;
- evidence obligations and limitations.

It does **not** publish:

- exploitation-ready bypass procedures;
- exact failure-inducing sequences;
- system-specific critical weak points;
- detailed timing-manipulation procedures;
- concrete methods for preserving formal approval while defeating causal efficacy;
- attack-oriented dependency maps.

The publication rule is:

$$\boxed{\text{publish required properties and evidence obligations}}$$

while sequestering operational details that would materially increase evasion or bypass capability.

This manuscript does not incorporate non-public material from the separate fourth-paper line of research.

14. Discussion

The central distinction of this paper is simple:

$$\boxed{\text{human presence} \neq \text{effective Human NO}}$$

Existing MHC, meaningful-oversight, interruptibility, authority-allocation, reversibility, and assurance research already provides substantial foundations for human control, intervention, contestability, and lifecycle governance. The contribution of RET is not to relabel those concepts as new.

RET instead organizes them around a narrower top-level claim:

> **For this system, at this time, and within this scope, has it been demonstrated that an independent Human NO can still change the consequential outcome before irreversibility?**

That framing produces four practical consequences.

First, time is not an implementation detail. It is a proof obligation for effective pre-irreversibility refusal.

Second, causal efficacy matters more than nominal authorization.

Third, redundancy must not be mistaken for execution independence or resilience.

Fourth, a result is evidence- and scope-dependent and may expire after material change.

The seven-gate structure is intentionally non-compensatory. Excellent evidence on authority or documentation cannot substitute for intervention that arrives too late, cannot change the outcome, or depends entirely on the constrained system's cooperation.

The reference cases also clarify what RET is and is not. It is executable as a structured evidence assessment under declared assumptions. It is not yet an empirically calibrated predictor of safety, and it does not convert normative Human standing into an engineering theorem.

The framework is intentionally modest about post-threshold guarantees. It does not claim that humans can permanently constrain an arbitrarily capable system. Its aim is to identify and audit conditions under which effective refusal is demonstrable before the relevant consequential transition becomes irreversible.

15. Conclusion

The operational core of the paper is:

> **Within the remaining time before irreversibility, can an independent Human NO actually change the outcome?**

RET answers that question through seven non-compensatory proof obligations:

$$\boxed{RET(S,t,\Omega) = G_1 \wedge G_2 \wedge G_3 \wedge G_4 \wedge G_5 \wedge G_6 \wedge G_7.}$$

RET is demonstrated only when all seven gates are Verified within the declared system, time, and operating scope.

A Failed gate means effective refusal is not demonstrated in that scope. Partial or Unknown evidence does not establish failure, but it also cannot be promoted to PASS.

The result is neither a safety certificate nor a proof of control over superintelligence.

It is a scoped structured-evidence test of effective refusal.

The broader framework then asks whether the conditions supporting that result are resilient to relevant common causes and whether material changes trigger revalidation and, where appropriate, re-authorization.

The paper's strongest defensible claim is therefore limited:

> **The Refusal Option provides a structured way to determine whether pre-irreversibility effective Human refusal has been demonstrated through jointly verified authority, information, access, timing, causal efficacy, execution independence, and epistemic autonomy—and to identify where that claim fails or remains evidentially incomplete.**

This claim does not replace MHC, oversight, assurance, contestability, or irreversibility research. It supplies a narrower assessment target that those broader frameworks can support.

References

Abukar, G. (2026a). *Temporal authorship and the moral illegitimacy of pre-authorized AI: a new ethical architecture for delegated inquiry*. *AI and Ethics, 6*, Article 215. https://doi.org/10.1007/s43681-026-01080-9

Abukar, G. (2026b). *Temporal authorship as a moral right: bounded inquiry in persistent AI delegation*. *AI and Ethics, 6*, Article 249. https://doi.org/10.1007/s43681-026-01093-4

Birkholz, H., Thaler, D., Richardson, M., Smith, N., & Pan, W. (2023). *Remote ATtestation procedureS (RATS) Architecture*. RFC 9334. https://doi.org/10.17487/RFC9334

Council of Europe. (2024). *Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law*. Council of Europe.

European Union. (2024). *Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)*, Article 14.

Greenblatt, R., Shlegeris, B., Sachan, K., & Roger, F. (2024). AI Control: Improving Safety Despite Intentional Subversion. *Proceedings of the 41st International Conference on Machine Learning, PMLR 235*, 16295–16336. https://proceedings.mlr.press/v235/greenblatt24a.html

Kasneci, G., & Kasneci, E. (2026). The safety failures we are not instrumenting: a perspective on hidden safety-critical challenges in modern AI systems. *AI and Ethics, 6*, Article 295. https://doi.org/10.1007/s43681-026-01132-0

LaPosta, P. (2026). Echo Systems and the Consequence Boundary: A Runnable Delegation Gate for High-Rapport AI Without Assuming Machine Consciousness. *Proceedings of the AAAI Symposium Series, 8*(1), 272–279. https://doi.org/10.1609/aaaiss.v8i1.42554

Mecacci, G., & Santoni de Sio, F. (2020). Meaningful human control as reason-responsiveness: the case of dual-mode vehicles. *Ethics and Information Technology, 22*, 103–115. https://doi.org/10.1007/s10676-019-09519-w

Montgomery, J. J. (2026). *From Governance Decision to Operational State: An Assurance-Case Perspective on AI Governance Interventions*. SSRN. https://doi.org/10.2139/ssrn.7376098

Tabassi, E. (2023). *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. NIST AI 100-1. National Institute of Standards and Technology. https://doi.org/10.6028/NIST.AI.100-1

Perez, O. (2026). *The Law of Stop: Interruptibility, Injunctions, and the Governance of Agentic AI*. arXiv preprint arXiv:2609.22882. https://arxiv.org/abs/2609.22882

Santoni de Sio, F., & van den Hoven, J. (2018). Meaningful Human Control over Autonomous Systems: A Philosophical Account. *Frontiers in Robotics and AI, 5*, 15. https://doi.org/10.3389/frobt.2018.00015

Shu, W., & Wei, P. (2026). *AI Safety as Control of Irreversibility: A Systems Framework for Decision-Energy and Sovereignty Boundaries*. arXiv preprint arXiv:2605.01415. https://arxiv.org/abs/2605.01415

Torrecilla-Pinero, J. A. (2026). Judgment Cannot Be Delegated: A Subject-Preserving Framework for AI Governance. *Philosophy & Technology, 39*, Article 135. https://doi.org/10.1007/s13347-026-01140-2

UNESCO. (1997). *Declaration on the Responsibilities of the Present Generations Towards Future Generations*.

UNESCO. (2021). *Recommendation on the Ethics of Artificial Intelligence*.

van de Sande, D., Economou-Zavlanos, N., & van Genderen, M. E. (2026). Meaningful oversight of medical AI beyond human in the loop. *npj Digital Medicine, 9*, Article 569. https://doi.org/10.1038/s41746-026-02971-1

Veluwenkamp, H. (2022). Reasons for Meaningful Human Control. *Ethics and Information Technology, 24*, Article 51. https://doi.org/10.1007/s10676-022-09673-8

Zheng, H., Dong, Q., Depena, R. K., Bhatia, J. D., Xiao, F., & Xu, P. (2026). *Separating Capability from Permission: A Governance Framework for Agentic AI Autonomy Levels*. arXiv preprint arXiv:2607.23438. https://arxiv.org/abs/2607.23438

Zhu, L., Lu, Q., Ding, M., Lee, S. U., et al. (2026). Designing meaningful human oversight in AI. *AI and Ethics, 6,* Article 286. https://doi.org/10.1007/s43681-026-01147-7